

17 Binomial Distributions

Example 17.1. [Biathlon](#) is a winter sport that combines cross-country skiing and rifle shooting. In Olympic sprint biathlon, each competitor competes in a ski race (10km for men, 7.5km for women) during which they also shoot at a total of 10 targets. Suppose that Campbell has a probability of 0.87 of successfully hitting any single target, independently.

Let X be the number of targets, out of 10, that Campbell successfully hits in their next sprint race.

1. Can the shots at the targets be considered Bernoulli trials?
2. What are the possible values of X ?
3. Describe in detail how you could use simulation to approximate the distribution of X .
4. Compute and interpret $P(X = 10)$.
5. Compute the probability that the Campbell makes their first 9 shots and misses the last.

6. Explain why the previous part is not $P(X = 9)$.
7. How many outcomes (S/F sequences of length 10) satisfy the event $\{X = 9\}$? Answer without listing all the possibilities.
8. Compute and interpret $P(X = 9)$.
9. Compute and interpret $P(X = 8)$.
10. Suggest a general formula for the probability mass function of X . Then use the pmf to create a table and spinner representing the distribution of X .
11. Suggest a simple shortcut formula for $E(X)$.
12. Compute and interpret $E(X)$. Did the shortcut formula work?

13. Compute $\text{Var}(X)$.

- Consider a *fixed number*, n , of Bernoulli(p) trials and let X count the number of successes. The distribution of X is defined to be the **Binomial(n, p) distribution**.
- A Binomial distribution is specified by two parameters:
 - n (an integer): the fixed number of Bernoulli trials, or the sample size
 - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
- A discrete random variable X has a **Binomial distribution** with parameters n , a nonnegative integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

- In the Binomial(n, p) situation, there are $\binom{n}{x}$ S/F sequences that result in exactly x successes—and thus $n - x$ failures—in n trials,
 - and each of these sequences has probability $p^x (1-p)^{n-x}$
- If X has a Binomial(n, p) distribution then

$$\begin{aligned} E(X) &= np \\ \text{Var}(X) &= np(1-p) \end{aligned}$$

Example 17.2. In each of the following situations determine whether or not X has a Binomial distribution. If so, specify n and p . If not, explain why not.

1. Roll a die 20 times; X is the number of times the die lands on an even number.
2. Roll a die 20 times; X is the number of times the die lands on 6.

3. Roll a die until it lands on 6; X is the total number of rolls.
4. Roll a die until it lands on 6 three times; X is the total number of rolls.
5. Roll a die 20 times; X is the sum of the numbers rolled.
6. Shuffle a standard deck of 52 cards (13 hearts, 39 other cards) and deal 5 *without replacement*; X is the number of hearts dealt. (Hint: be careful about why.)
7. Roll a fair six-sided die 10 times and a fair four-sided die 10 times; X is the number of 3s rolled (out of 20).

Example 17.3. Xavier bets on red on each of 3 spins of a roulette wheel. After the third spin, Xavier leaves and Zander arrives and bets on black on the next two spins. Let X be the number of bets that Xavier wins and let Z be the number that Zander wins.

1. Identify the distribution of X and the distribution of Z .
2. Are X and Z independent? Explain.
3. Compute $P(X = 2, Z = 1)$.

